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How Point of Care Oral Cancer Biosensing Diagnostics can make the World a Better Place

There is a famous Latin phrase that many healthcare professionals and engineers in medicine knows or at least the English version by heart, *primum non nocere* or “do no harm”. Many of those who go into the healthcare industry or biomedical engineering want to help people by making their lives a little easier. Medical devices and treatments come in all sorts of shapes, sizes, forms, and functions. They range from simple medical screws, to super complicated surgical robots and to advanced diagnostic medical AI tools. They all try to restore biological function, augment human movement, or prevention. If the job of an engineer is to design and build tools that improve human life then the things we design should also make society and by extension the world a better place. As Dr. Gregory Courand from his *Deploying Technology to Advance Justice* paper puts it, “a good society detects and addresses forms of inequality that are associated with undesirable or negative conditions via righting wrongs and implementing solutions going forward”. Unfortunately our society does not actually reflect this current sentiment. Access to medical devices and treatment is not equal as evidenced by a study from the World Health Organization who found that people in lower socioeconomic groups consistently deal with severe financial roadblocks that lead to delayed care and much worse survival outcomes. On top of socioeconomic class divides there is ample evidence that different

racial groups do not get the same access to quality care. As Courand points out also from his paper, "the empirical reality is that people of color, and the poor, are less likely to be well educated and healthy than white people of some means". This is reinforced from the Centers for Disease Control and Prevention (CDC 2024) a U.S. government organization via their report on minority health which reports that racial and ethnic minority groups across the United States face much higher rates of preventable illness and death across the board (“Racism and Health”). It is clear to see the overall picture that with society having negative conditions along with systemic inequality there is a clear case of injustice that puts vulnerable groups in unfair, prejudicial, and anti-flourishing conditions as described in the course readings.

The reasons behind healthcare injustice are many and complicated but a few come to mind. Vulture capitalism, corporate mindset of profits over patient care, and resource scarcity are the big ones. The primary actors perpetuating these harms are medical device companies, insurance companies, and in certain cases government institutions via uneven medicaid reimbursement rates and local tax-base hospital funding schemes (Yearby 520). When someone is sick, they frequently do not have the luxury of shopping around in a fair, free market. Alongside corporate greed, pervasive systemic issues like structural racism and classism keep this broken system in place (Courand 10). Because hospital funding and private investment usually flow into wealthy neighborhoods, rich areas get state-of-the-art diagnostic labs while low-income communities deal with constant resource scarcity (Courand 5). As long as medical device companies and insurance companies are obsessed with maximizing profits for shareholders, these inequalities are not going anywhere. We see this play out constantly in cancer diagnostics. Catching cancer early typically is done via invasive tissue biopsies, specialist visits,

and expensive lab work. A study in the *Journal of Clinical Oncology* shows how the massive out-of-pocket costs and complex hospital infrastructure required for tissue biopsies and advanced diagnostics cause severe "financial toxicities," which prices out uninsured and low-income patients until their cancer has already spread (de Souza et al. 12).

To combat this injustice, my senior design team is working on a low-cost, point-of-care (PoC) diagnostic device for oral cancer. We are trying to build the equivalent of an at-home COVID-19 rapid test, except our sensor detects oral cancer by measuring an oral cancer biomarker called Matrix Metalloproteinase-9 (MMP-9) in saliva. Our device uses screen-printed gold electrodes and a technique called electrochemical impedance spectroscopy (EIS) to detect when MMP-9 proteins bind to antibodies on the sensor's surface. The goal is to have the sensor give a clear electrical reading in a matter of minutes. Instead of forcing patients to go through painful, invasive biopsies or long fluid extractions right away, this device would let local community healthcare providers and eventually patients themselves to test their own saliva via over the counter devices for early signs of oral cancer quickly and painlessly.

For us building the device is a direct way to fix a broken power dynamic in healthcare in addition to being a cool technical challenge. Big medical corporations, insurance companies, and hospital specialists hold all the cards most of the time in a traditional treatment relationship. Because standard oral cancer testing requires a surgeon to cut out a piece of tissue and a pathology lab to stain and read it, the medical system has total control over who gets tested based on who can pay and who has good insurance. Dr. Courand classifies this kind of institutional gatekeeping as an anti-flourishing condition because it places strict "limitations on self-determination... owing to limitations on choices and barriers to advancement, access, and

resources" (Courand 3). When patients cannot afford to see a specialist just to check a suspicious mouth lesion, their freedom and agency to manage their own health is taken away. By giving agency and accessible options directly to the patient and local community clinics, we empower people to make informed decisions about their own healthcare before it is too late. Our proposed salivary biosensor does exactly that by taking the first step of cancer screening out of expensive hospital labs and putting it directly into the hands of the public.

To actually advance justice, a technology has to asymmetrically help the people who are suffering the most. As discussed earlier, the worst outcomes of oral cancer are asymmetrically experienced by low-income and minority populations who cannot afford early biopsies. The best part about our point of care device is that it specifically benefits these vulnerable populations the most. Wealthier, more affluent patients are going to keep using traditional tissue biopsies and big hospital labs. They already have the money and insurance coverage to get the gold-standard treatment without worrying it affecting their wallets as much. On the other hand, our oral cancer biosensor asymmetrically benefits populations with fewer resources. Academic studies on global health have repeatedly proven that cheaper, faster point-of-care tools are used much more heavily by poorer, underserved populations who are locked out of standard laboratory care, leading to massive drops in mortality for those specific groups (Drain et al. 220).

Stepping back and looking at the bigger picture, taking account of the course material's concepts like negative and asymmetric conditions and actively looking for ways to create justice is going to completely change how I approach my future career as an engineer. There is a great quote from the end of Dr. Courand's paper, "As engineers, we are called to do more than simply create things with benefits for some; we should at least consider creating socially therapeutic

technologies so that, incrementally, the world becomes more just" (15). Going into the medical device industry, I really want to focus on a utilitarian view of bettering society. In so many of our engineering classes, we are trained to focus on function and cost alone. We spend months worrying about whether our CAD tolerances are tight enough, whether our Python scripts run fast enough, or whether our device cost is minimized. Those technical details obviously matter, because a medical device cannot help anyone if it does not work. However, if I only care about technical specs and manufacturing margins, I am just going to end up building expensive toys for rich hospitals, which only perpetuates the injustice. As such, everything I do in the near future as a biomedical engineer needs to keep injustice in mind. Whenever I am working on design changes, picking out materials, or helping push a product to release, I hope to build constraints into my work that specifically protect underserved populations. Examples can be keeping unit costs low enough for free clinics and making sure a device does not require a million-dollar lab to run. At the end of the day, being a good engineer means caring about the kind of society we are helping to build. As Sandel writes from the course reading paper, "In times of trouble, a good society pulls together. Rather than press for maximum advantage, people look out for one another. A society in which people exploit their neighbors for financial gain in times of crisis is not a good society" (Sandel 7). Getting sick with cancer or disease is one of the worst situations a person can be in. By having the mindset of choosing human lives over corporate profits and designing tools that actually reach the people who need them most, I can hopefully make sure I leave the world a better place.

Works Cited

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